

Executable Reproduction of Equation (2.2) in the OpenAI Erdős Unit-Distance Disproof Remarks

A High-Precision Python Check of the Published Eq. (2.2) Calculation

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Version 0.2.5 — 2026-05-25

Abstract

We present `openai-erdos-eq22-reproduction`, an open-source Python package that re-derives, in pure NumPy and `mpmath`, the finite and numerically explicit parts of the 2026 disproof of the Erdős planar unit-distance conjecture announced by an OpenAI reasoning model [Ope26] and written up by Alon, Bloom, Gowers, Litt, Sawin, Shankar, Tsimmerman, Wang, and Matchett Wood [Alo+26]. The central object is equation (2.2) of the remarks paper: for the explicit choice $T = \{3, 5, 7, 11, 13, 17\}$, $S = \{101, \infty\}$, and $L_T = \mathbb{Q}(\sqrt{5}, \sqrt{13}, \sqrt{17}, \sqrt{21}, \sqrt{33})$, that equation states a rigorous exponent excess of approximately 6.24×10^{-38} above the classical Erdős exponent. Our `mpmath` evaluation at 200 bits of precision returns 6.2391×10^{-38} , matching the published two-significant-figure value to 1.4×10^{-4} relative error.

We make no other mathematical claim. The artifact does *not* re-prove Theorem 1.1 of [Alo+26], does *not* construct the infinite Golod–Shafarevich tower whose existence is quoted as an input to that theorem, and does *not* address any separately published sharpening of the bound (in particular, it does *not* reproduce or verify the explicit value $\delta \approx 0.014$ that appears in Sawin’s separate explicit-lower-bound preprint or in informal coverage of the result). The intended audience is anyone who, having read §2 of [Alo+26], wants a citable, CI-tested Python implementation of its eq. (2.2) on which they can run their own sensitivity checks. The package is distributed under the MIT licence at <https://github.com/Flamehaven-Labs/openai-erdos-eq22-reproduction>.

Reproduction. The repository provides a one-command verifier and frozen JSON/Markdown evidence files; the exact commands and evidence paths are given in §7. Per-release Zenodo DOIs and citation metadata are maintained in `CITATION.cff` and on the [GitHub Releases](#) page, not inlined here; this keeps the manuscript surface stable across patch releases.

*Independent author. Code distribution: github.com/Flamehaven-Labs/openai-erdos-eq22-reproduction.

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1 Background and scope

1.1 The Erdős unit-distance problem

For a finite set $P \subset \mathbb{R}^2$, let $\nu(P) = \#\{\{x, y\} \subset P : |x - y| = 1\}$ and $\nu(n) = \max_{|P|=n} \nu(P)$. The planar unit-distance problem [Erd46] concerns the asymptotic behaviour of $\nu(n)$. Erdős proved $\nu(n) \geq n^{1+\Omega(1/\log \log n)}$ via a square-grid construction and conjectured the matching upper bound $\nu(n) \leq n^{1+C/\log \log n}$ for some absolute constant C .

1.2 The 2026 disproof

The conjecture was disproved in May 2026 by an internal reasoning model at OpenAI [Ope26], with a simplified construction and explicit numerical lower bound published the following day by Alon, Bloom, Gowers, Litt, Sawin, Shankar, Tsimmerman, Wang, and Matchett Wood [Alo+26]. The combined result is:

Theorem 1.1 of [Alo+26]. *There exists an absolute constant $\delta > 0$ and infinitely many positive integers n such that $\nu(n) \geq n^{1+\delta}$.*

The construction passes through a CM field $K_j = L_j(i)$, where the L_j form an infinite tower of totally real fields of bounded root discriminant, supplied by the Golod–Shafarevich theorem [GS64]. Section 2 of [Alo+26] gives an explicit choice with $T = \{3, 5, 7, 11, 13, 17\}$, $S = \{101, \infty\}$, and $L_T = \mathbb{Q}(\sqrt{5}, \sqrt{13}, \sqrt{17}, \sqrt{21}, \sqrt{33})$, and gives (eq. (2.2)) a rigorous lower bound whose exponent excess is approximately 6.24×10^{-38} .

1.3 What this artifact reproduces

This paper accompanies a Python package, `openai-erdos-eq22-reproduction`, whose scope is deliberately narrow:

- we *do not* re-prove Theorem 1.1 of [Alo+26];
- we *do not* construct the infinite Golod–Shafarevich tower whose existence is invoked in the proof;
- we *do not* verify any separately published sharpening of eq. (2.2), including but not limited to Sawin’s explicit-lower-bound preprint and any informal value such as $\delta \approx 0.014$ circulating in secondary coverage;
- we *do* re-implement, in pure Python, the finite and explicitly computable steps of the eq. (2.2) construction: small-integer arithmetic for the algebraic-number-theory layer and `mpmath` at 200-bit precision for the final numerical evaluation;
- we *do* pin the published value $\delta_{\text{excess}} \approx 6.24 \times 10^{-38}$ as a regression test that fails if our code drifts from the source.

The artifact is therefore neither a formal proof checker (it does not discharge proofs through a trusted logical kernel such as Lean or Coq) nor an independent mathematical contribution (it produces no new theorem and no improved bound). It is a reproducibility artifact in the spirit of an ACM Artifact Evaluation submission: a citable, version-pinned, CI-tested executable that lets a third party re-derive the published numerical value on their own hardware, without trusting our reading of the source PDFs.

The remainder of the paper describes the three implementation phases (§§2–4), the central numerical reproduction of eq. (2.2) (§5), the explicit limits (§6), and the reproducibility surface (§7).

2 Phase 1: faithful $h(K) = 1$ lattices

Phase 1 implements the trivial-class-number degenerate case of Lemma 2.2 of [Ope26]: an imaginary quadratic field $K = \mathbb{Q}(\sqrt{-d})$ with class number $h(K) = 1$, where the ideal-class pigeonhole reduces to the identity. For these fields the ring of integers $\mathcal{O}_K$ embeds standardly in $\mathbb{R}^2$ as a rank-2 lattice, and unit distances between lattice points can be counted exactly.

2.1 Coverage

We implement the three smallest $h = 1$ cases (the Baker–Heegner–Stark list 1, 2, 3, 7, 11, 19, 43, 67, 163 is encoded in the module-level constant `CLASS_NUMBER_ONE_DISCRIMINANTS`; the first three are implemented):

d	$\mathcal{O}_K$	Standard embedding into $\mathbb{R}^2$
1	$\mathbb{Z}[i]$ (Gaussian integers)	$a + bi \mapsto (a, b)$
2	$\mathbb{Z}[\sqrt{-2}]$	$a + b\sqrt{-2} \mapsto (a, b\sqrt{2})$
3	$\mathbb{Z}[\omega], \omega = \frac{-1+\sqrt{-3}}{2}$	$a + b\omega \mapsto (a - b/2, b\sqrt{3}/2)$

The later Baker–Heegner–Stark cases are not needed for the eq. (2.2) reproduction and introduce no new mechanism for this artifact: they change the quadratic basis but not the finite embedding/counting pattern used here. We therefore keep Phase 1 as a minimal exact reference rather than a catalogue of all class-number-one quadratic fields.

For each d and bound $b \geq 0$, the module enumerates all $(2b+1)^2$ lattice points with integer coefficients in $[-b, b]$ and counts unit-distance pairs exactly via a numpy $O(n^2)$ broadcast.

2.2 Verifications

Reproduction 2.1. For $\mathbb{Z}[i]$ with $b = 1$ (the 3×3 Gaussian grid, $n = 9$), the artifact computes exactly 12 unit-distance pairs: three rows of two horizontal edges plus three columns of two vertical edges. Pinned in `tests/test_imaginary_quadratic_lattice.py : test_gaussian_3x3_grid_has_12_unit_distances`.

Reproduction 2.2. For $\mathbb{Z}[i]$ with $b = 3$ (the 7×7 Gaussian grid, $n = 49$), the artifact computes exactly $2 \cdot 7 \cdot 6 = 84$ unit-distance pairs. Source: `src/erdos_ant/imaginary_quadratic_lattice.py`, class `ImaginaryQuadraticLattice.count_unit_distance_pairs`. Pinned in `src/erdos_ant/verify.py` as the assertion `phase1_gaussian_7x7_grid_has_expected_84_pairs`.

Reproduction 2.3. For each $b \in \{1, 2, 3, 4\}$, the Eisenstein-lattice (triangular) unit-distance count strictly exceeds the Gaussian-lattice count at the same b . The triangular lattice has 6 nearest neighbours per interior point versus the square lattice’s 4. Pinned in `test_eisenstein_unit_distances_exceed_gaussian_for_same_bound`.

2.3 Non-claims

Phase 1 does not improve the asymptotic Erdős exponent. With $[K : \mathbb{Q}] = 2$, the construction is the classical square or triangular grid; in particular $\nu(n) \leq n^{1+O(1/\log \log n)}$ holds and the $\delta > 0$ excess is not present. The asymptotic result strictly requires $[K : \mathbb{Q}] \rightarrow \infty$, which is the Phase 3 setting.

Phase 1 is included in the artifact for two reasons: (i) it is a self-contained, exact reference against which the Phase 3 construction can be sanity-checked at low parameter values, and (ii) it documents the standard complex embedding of $\mathcal{O}_K$ in $\mathbb{R}^2$ unambiguously, which is the substrate of every later step.

3 Phase 2: $K = \mathbb{Q}(\sqrt{-5})$ with genus theory

Phase 2 implements the smallest non-trivial-class-number case $K = \mathbb{Q}(\sqrt{-5})$, $h(K) = 2$. The Lemma 2.2 pigeonhole of [Ope26] becomes non-vacuous here: when an ideal product $A_\varepsilon = \prod_j P_j^{\varepsilon_j} (P'_j)^{k-\varepsilon_j}$ lands in the non-principal class, no element $\alpha_\varepsilon \in K^*$ with $(\alpha_\varepsilon) = A_\varepsilon A_\eta^{-1}$ exists for arbitrary η — the pigeonhole must restrict to a single ideal class, and the bound $|U| \geq (k+1)^s/h(K)$ is sharp.

3.1 Classical facts encoded

- $\mathcal{O}_K = \mathbb{Z}[\sqrt{-5}]$ with discriminant -20 (K_DISCRIMINANT = -20, CLASS_NUMBER = 2).
- The non-trivial ideal class is represented by $\mathfrak{p}_2 = (2, 1 + \sqrt{-5})$, satisfying $\mathfrak{p}_2^2 = (2)$.
- By genus theory [Cox89, Theorem 6.1], the Hilbert class field is the genus field $H = K(\sqrt{-1}) = \mathbb{Q}(i, \sqrt{5})$.
- A rational prime p (odd, $p \neq 5$) splits completely in K iff $\left(\frac{-5}{p}\right) = 1$, equivalently $p \bmod 20 \in \{1, 3, 7, 9\}$. Of these, $p \bmod 20 \in \{1, 9\}$ gives a principal prime (representable as $a^2 + 5b^2$), and $p \bmod 20 \in \{3, 7\}$ gives a non-principal prime.

3.2 Lemma 2.2 pigeonhole, explicit

The Phase 2 module `erdos_ant.genus_class_field` exposes a `GenusClassFieldProbe` that, given a tuple of split rational primes, computes the ideal-class partition of the $\{0, 1\}^s$ family of ideal products and reports $|U| \geq (k+1)^s/h(K)$.

Reproduction 3.1. For the choice $(p_1, p_2) = (29, 41)$ (both in the principal genus, $29 = 3^2 + 5 \cdot 2^2$, $41 = 6^2 + 5 \cdot 1^2$), all four candidate ideals A_ε for $\varepsilon \in \{0, 1\}^2$ are principal, and the Lemma 2.2 lower bound is $(1+1)^2/2 = 2$. Pinned in `tests/test_genus_class_field.py::test_lemma_22_pigeonhole_with_two_principal_primes`.

Reproduction 3.2. For the choice $(p_1, p_2) = (29, 3)$ (one principal, one non-principal), the class group has order 2, so the prime above 3 and its conjugate lie in the same non-principal class. Consequently all four ideals land in the non-principal class rather than splitting $2+2$ between the two classes, and the Lemma 2.2 lower bound is still 2. Pinned in `test_lemma_22_pigeonhole_with_mixed_genus`.

3.3 Non-claims

Phase 2 fixes a single CM field of degree 2 over $\mathbb{Q}$ and produces a finite pigeonhole demonstration. It does not produce a sequence of fields with $[K_j : \mathbb{Q}] \rightarrow \infty$, and does not generate the exponentially growing set U that drives the asymptotic exponent excess. It does, however, instantiate the smallest concrete example where the pigeonhole step is genuinely non-trivial.

4 Phase 3: the Sawin multi-quadratic construction, finite parts

Phase 3 implements the finite, explicitly checkable parts of the *simplified* construction of Section 2.1 of [Alo+26], which uses an unramified pro-2 tower over a multi-quadratic base field rather than the original GPT pro-3 tower over a cyclic cubic. The simplification is due to Sawin and yields the explicit numerical bound evaluated in §5.

4.1 The Sawin parameters, verbatim

The construction fixes

$$\begin{aligned} T &= \{3, 5, 7, 11, 13, 17\}, \\ S &= \{101, \infty\}, \\ L_T &= \mathbb{Q}(\sqrt{5}, \sqrt{13}, \sqrt{17}, \sqrt{21}, \sqrt{33}), \end{aligned}$$

with $[L_T : \mathbb{Q}] = 2^5 = 32$, and sets $K = L_T(i)$ as the CM field.

These values appear verbatim in the artifact:

- T_SET = (3, 5, 7, 11, 13, 17)
- S_SET_SPLIT = (101,) (the infinite place is implicit)
- L_T_GENERATOR_RADICANDS = (5, 13, 17, 21, 33)
- SAWIN_R_PRODUCT = 2 · 3 · 5 · 7 · 11 · 13 · 17 = 510510

and the test suite pins each constant against the published source.

4.2 Splitting of 101 in L_T

A rational prime p splits completely in $\mathbb{Q}(\sqrt{d_1}) \cdot \mathbb{Q}(\sqrt{d_2}) \cdots \mathbb{Q}(\sqrt{d_s})$ if and only if it splits completely in each factor $\mathbb{Q}(\sqrt{d_i})$ (which, for linearly disjoint quadratic extensions, is the definition of completely splitting in the compositum). For odd $p \neq d_i$, splitting in $\mathbb{Q}(\sqrt{d_i})$ is equivalent to $\left(\frac{d_i}{p}\right) = 1$.

Reproduction 4.1. *The artifact evaluates the Jacobi symbol $\left(\frac{d_i}{101}\right)$ for each $d_i \in \{5, 13, 17, 21, 33\}$ and finds all five symbols equal to +1. Therefore 101 splits completely in L_T . Source: `src/erdos_ant/sawin_multiquadratic.py`, function `splits_completely_in_L_T`. Pinned in `tests/test_sawin_multiquadratic.py::test_101_splits_completely_in_each_Q_sqrt_d`.*

4.3 Galois-rank counts and Golod–Shafarevich admissibility

Let $G = \text{Gal}(\tilde{\mathbb{Q}}_T/\mathbb{Q})$ be the Galois group of the maximal pro-2 extension of $\mathbb{Q}$ unramified outside T . Because T contains a prime that is 3 (mod 4) (e.g. 3, 7, or 11), the source argument quotes the corresponding genus-theoretic rank count as $|T| - 1 = 5$, giving

$$d(G_T^\infty) = |T| - 1 = 5.$$

The artifact records and tests this finite rank count; it does not reprove the class-field-theoretic rank formula. Adjoining S adds at most $|S| - 1 = 1$ Frobenius relations (Sawin’s pro-2 analogue of the Hajir–Maire–Ramakrishna cut), so

$$r(G_T^S) \leq d(G_T^\infty) + |S| - 1 = 5 + 1 = 6.$$

The Golod–Shafarevich admissibility condition for the tower to remain infinite is $r \leq d^2/4$; here

$$r(G_T^S) \leq 6 \leq \frac{(|T| - 1)^2}{4} = \frac{25}{4} = 6.25,$$

with positive finite margin 0.25. By the cited Golod–Shafarevich theorem [GS64], this admissibility implies the tower is infinite. The artifact verifies the admissibility inequality only; it does *not* itself prove or compute the infinitude — the cited theorem does.

Reproduction 4.2. *The artifact computes $d(G_T^\infty) = 5$, $r(G_T^S) \leq 6$, and the Golod–Shafarevich threshold $25/4 = 6.25$, and confirms $6 \leq 6.25$. Pinned in `test_d_G_infty_equals_T_minus_1_because_T_contains_3_mod_4`, `test_r_bound_is_six_per_remarks_pdf`, and `test_golod_shafarevich_inequality_holds`.*

4.4 Non-claims

Phase 3 does *not* construct the infinite tower itself. The existence of the tower is a non-constructive consequence of the Golod–Shafarevich theorem [GS64] together with Sawin’s pro-2 analogue of the Hajir–Maire cut [HM01; Alo+26]; no finite computation can exhibit a closed-form description of L_j for arbitrary j . Phase 3 checks every step of the construction that *is* amenable to finite computation, and stops at that boundary.

5 Numerical reproduction of equation (2.2)

This is the central numerical content of the artifact. We reproduce equation (2.2) of [Alo+26] as a regression test against the published source.

5.1 The published expression

Sawin et al. state immediately after Lemma 2.1 [Alo+26, §2.1, eq. (2.2)]:

Explicitly, the exponent in (2.1) can be taken to be

$$1 + \frac{\log(u\pi/(36v))}{\log(36/\delta^2)} \approx 1 + 6.24 \cdot 10^{-38}$$

where $v = r/2$, $\delta \geq 101^{-2\lceil 18r^3/\pi \rceil}$, and $u = \lceil 18r^3/\pi \rceil \cdot r^{-2}$, with $r = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17$.

Let $K = \lceil 18r^3/\pi \rceil$, so that $u = K/r^2$ and $\delta = 101^{-2K}$. Direct substitution gives

$$\frac{u\pi}{36v} = \frac{(K/r^2) \cdot \pi}{36 \cdot (r/2)} = \frac{K\pi}{18r^3},$$

and the exponent excess above 1 simplifies to

$$\delta_{\text{excess}} = \frac{\log(K\pi/(18r^3))}{\log 36 + 4K \log 101}. \quad (1)$$

5.2 Why float64 fails

The numerator of (1) is $\log(K\pi/(18r^3))$. By definition of the ceiling, $K \geq 18r^3/\pi$, so the ratio $K\pi/(18r^3)$ lies in the half-open interval

$$\left(1, 1 + \frac{\pi}{18r^3}\right].$$

For $r = 510510$, $r^3 \approx 1.33 \cdot 10^{17}$, hence $\pi/(18r^3) \approx 1.31 \cdot 10^{-18}$.

The numerator is therefore $\log(1 + \varepsilon)$ for some $\varepsilon \in (0, 1.31 \cdot 10^{-18}]$. IEEE-754 `float64` has a 52-bit mantissa and machine epsilon $\approx 2.22 \cdot 10^{-16}$. The computation also forms $K\pi$ with $K \approx 7.62 \cdot 10^{17}$ before dividing by $18r^3$ to obtain a ratio extremely close to 1. At ordinary double precision the near-one ratio is rounded at a scale larger than the 10^{-18} separation, so the cancellation happens before the logarithm: the naive evaluation rounds the ratio to exactly 1, yields $\log 1 = 0$, and reports $\delta_{\text{excess}} = 0$.

This is not a flaw in eq. (1); it is a property of finite-precision arithmetic. Any honest evaluation must use a precision significantly above 1 part in 10^{18} .

5.3 Evaluation in mpmath at 200-bit precision

The artifact uses mpmath [The23] with `mp.prec = 200` (about 60 decimal digits) to evaluate the quantities in eq. (1) exactly enough that the $1 + 10^{-18}$ separation survives. The function `evaluate_sawin_exponent_bound` in `erdos_ant.sawin_multiquadratic` returns a `SawinDeltaEvaluation` dataclass containing r , K , v , $\log u$, and δ_{excess} .

The 200-bit precision is a conservative budget, not an unexplained output target. Resolving a $1.31 \cdot 10^{-18}$ numerator separation requires about $\log_2(1/(1.31 \cdot 10^{-18})) \approx 59.4$ bits just to see the signal; keeping the relative error in that small numerator below the 10^{-3} regression threshold adds about 10 guard bits. The chosen 200 bits leave a large deterministic margin over this rough 70-bit lower bound while remaining fast enough for CI.

Reproduction 5.1. *Calling `evaluate_sawin_exponent_bound()` (source: `src/erdos_ant/sawin_multiquadratic.py`, function `evaluate_sawin_exponent_bound`) returns*

$$r = 510,510, \quad K = \lceil 18 \cdot 510510^3 / \pi \rceil \approx 7.62 \cdot 10^{17},$$

and

$$\delta_{\text{excess}} = 6.2391 \cdot 10^{-38}.$$

The published value is $\approx 6.24 \cdot 10^{-38}$ (given to two significant figures in [Alo+26]). The relative error between the artifact's evaluation and the published value is

$$\frac{|6.2391 - 6.24|}{6.24} = 1.4 \cdot 10^{-4} = 0.014\%,$$

fully consistent with two-significant-figure rounding in the source (see also `docs/DEVIATION_LOG.md` for the full deviation ledger). Pinned in `tests/test_sawin_multiquadratic.py::test_sawin_exponent_bound_matches_remarks_eq_2_2`. The per-source-file SHA-256 manifest under which this number was computed appears in `reports/verification_result.json` field `source_sha256_manifest`.

5.4 What the verification establishes (and does not)

Verification 5.1 establishes that, given the inputs T, S, L_T, K exactly as written in [Alo+26, §2.1], the elementary closed-form expression eq. (1) evaluates to the value the source paper claims, to within published rounding.

This does *not* establish:

- that the closed-form expression is itself correct (that depends on Lemma 2.1 and earlier algebra of [Alo+26], which we do not re-derive);
- that the underlying infinite Golod–Shafarevich tower exists (Phase 3 only verifies the finite admissibility condition);
- that $\delta_{\text{excess}} = 6.24 \cdot 10^{-38}$ is in any sense optimal — it is one explicit lower bound, not the best possible.

What it does establish is that any future code change in the artifact that drifts away from the published value will fail the regression test at the 10^{-3} relative-tolerance band hard-coded in the test. The observed relative error is $1.4 \cdot 10^{-4}$, i.e. 0.014%; the threshold is 10^{-3} , i.e. 0.1%.

6 Limits and explicit non-claims

This section states the boundary of the artifact before the reproducibility details. The goal is not to diminish the reproduction, but to keep the numerical result, proof input, and peer-review status separate.

6.1 This artifact does NOT contain a new proof

Theorem 1.1 of [Ope26; Alo+26] is proved by those authors. The present paper does not re-derive Lemma 2.1, Lemma 2.2, Proposition 2.3, or any other intermediate result. Where the source papers say “by the Golod–Shafarevich theorem” or “by the Hajir–Maire tower-cutting argument”, we likewise take those classical results as given.

6.2 This artifact does NOT construct the infinite tower

The existence of L_j with $[L_j : \mathbb{Q}] \rightarrow \infty$ and bounded root discriminant is supplied by Golod–Shafarevich [GS64] together with the Hajir–Maire splitting technique [HM01] adapted to pro-2 in [Alo+26]. This is a non-constructive existence theorem. No finite computation can exhibit the entire infinite tower. Phase 3 of the artifact stops at the boundary of computable admissibility.

6.3 This artifact has NOT been peer-reviewed

No external mathematician has reviewed this code base. The grounds on which a reader might choose to take the artifact’s output at face value are the following reproducible evidence surfaces — none of which is a substitute for peer review:

1. the 60-test unit suite passes on CI (Ubuntu + Windows $\times$ Python 3.11 / 3.12);
2. the central numerical reproduction in §5 matches the published value $\approx 6.24 \cdot 10^{-38}$ to a relative error of $1.4 \cdot 10^{-4}$, consistent with the two-significant-figure rounding in the source;
3. the source is small ($\approx 1\,000$ lines across four modules) and intended to be readable cover-to-cover by anyone who has read §2 of [Alo+26].

We treat the artifact as input to mathematician review, not as a substitute for it.

6.4 Reported 0.014 and 0.0318 are NOT in either formal PDF

Secondary coverage (e.g. [Kal26]) reports numerical values $\delta \approx 0.014$ and $\delta \approx 0.0318$ in the informal contexts $n^{1.014}$ and $n^{1.0318}$. We verified by direct text extraction with `pypdf` that neither of these literal numerical values appears in either [Ope26] or [Alo+26]. The only rigorous numerical lower bound that appears verbatim in either formal PDF is $\delta_{\text{excess}} \approx 6.24 \cdot 10^{-38}$ from eq. (2.2) of [Alo+26]. This is a text-search and source-review claim, not a formal statement about every rendered glyph in the PDFs.

The artifact records this three-way provenance in a small registry of tagged `SOURCE_CLAIMS` entries:

- `openai_index_reported` ($\delta = 0.014$), flagged as not literally present in either formal PDF;
- `sawin_rigorous_lower_bound` ($\delta = 6.24 \cdot 10^{-38}$), flagged as the only numerical bound appearing verbatim in [Alo+26];
- `sawin_refinement_reported` ($\delta = 0.0318$), flagged as not literally present in either formal PDF.

Reproducing 0.014 or 0.0318 would require a different artifact working from a different (and in the first case unpublished) source; that is outside the scope claimed here.

6.5 This artifact does NOT improve the published bound

We made no attempt to optimise the choice of T , S , L_T or the inner parameter k to obtain a larger δ_{excess} . The explicit choice of [Alo+26, §2.1] is what we encode and reproduce; the optimisation problem of finding (T, S, k) with larger admissible δ remains open and is out of scope.

7 Reproducibility

7.1 One-line reproduction

```
git clone https://github.com/Flamehaven-Labs/openai-erdos-eq22-reproduction
cd openai-erdos-eq22-reproduction
pip install -e ".[dev]"
python -m erdos_ant.verify
```

Expected terminal output:

```
Verdict: PASS
Checks: 21/21 passed
eq (2.2) exponent excess: 6.2391e-38 (published ~6.24e-38, rel.err 0.0001)
(frozen-report mode: tracked evidence not written; pass -write-evidence
to refresh)
```

By default the command does not write to disk; the working tree stays clean. Release maintainers pass `-write-evidence` to refresh the tracked files `reports/verification_result.json` and `reports/verification_report.md`.

The line `Checks: 21/21 passed` refers to the verifier’s top-level checklist. One of those checks is a subprocess execution of the 60-test pytest suite with external pytest plugin autoloading disabled; running `pytest -q` directly executes the same unit tests outside the verifier wrapper.

7.2 Pinned environment

Item	Value
Repository	https://github.com/Flamehaven-Labs/openai-erdos-eq22-reproduction
Version (paper source)	v0.2.5
Python	3.11 and 3.12 (CI-verified)
numpy	≥ 1.26
mpmath	≥ 1.3
Test count	60 unit tests + 21 verifier checks
CI matrix	Ubuntu $\times$ Windows $\times$ Python 3.11 / 3.12
Licence	MIT
Per-release DOIs	CITATION.cff and GitHub Releases

7.3 Supplementary local checks

In addition to the unit suite and verifier, the release process keeps author-operated local source-level checks for claim framing, traceability, and repository hygiene. Those checks are recorded as engineering evidence only. They are not independent review, are not mathematical peer review, and are not used as a premise for the correctness of the eq. (2.2) reproduction. See §6 for the bound on what local release checks can and cannot attest to.

7.4 Citation

A `CITATION.cff` file in the repository root provides citation metadata in the Citation File Format v1.2 standard, recognised by Zenodo, GitHub, and most citation managers. The recommended citation for a specific tagged release is the Zenodo DOI associated with that tag.

Acknowledgements

We thank the OpenAI reasoning model whose May 2026 output prompted this reproduction effort, and the authors of the companion remarks paper [Alo+26] — Noga Alon, Thomas F. Bloom, W. T. Gowers, Daniel Litt, Will Sawin, Arul Shankar, Jacob Tsimerman, Victor Wang, and Melanie Matchett Wood — for the explicit choice of parameters in their §2 that made a finite numerical reproduction tractable in the first place.

We thank Farshid Hajir, Christian Maire, and Ravi Ramakrishna for the class-field tower-cutting technique [HM01] that is invoked, but not re-derived, in this artifact’s Phase 3; and David A. Cox, whose textbook treatment of genus theory underlies Phase 2.

This artifact was developed and released by the author independently. The supplementary local checks listed in §7 were run as part of the release process; their outputs are treated as recorded engineering evidence, not as independent mathematical endorsement.

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